higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PLUMBING
NQF LEVEL 4

(12030544)

12 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 5 pages.

NCV

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Answer ALL questions in the ANSWER BOOK.
5. Start each question on a NEW page.
6. Sketches should be neat, labelled and large enough so that ALL details are clear.
7. ALL abbreviations and symbols must comply with the latest National Building Regulations and ALL relevant SANS (SABS) codes.
8. Write neatly and legibly.

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QUESTION 1: BASIC BUILDING WORK AND BUILDING A MANHOLE

1.1 Name TWO commonly used tools to check foundations. (2)

1.2 Which TWO types of joints are generally used to fit the parts of a precast-concrete manhole together? (2)

1.3 What is an appropriate description of "*benching*"? (2)

1.4 Why are certain pipes used in plumbing jobs called "*half channels"*? (2)

1.5 Give TWO purposes of a valve boxes. (2)
[10]

QUESTION 2: BELOW-GROUND DRAINAGE SYSTEMS

2.1 Which TWO levels should be used when laying underground drainage? (2)

2.2 List FOUR important procedures we need to follow during the setting-out, and excavation of trenches? (4)

2.3 Name FIVE levelling instruments that can be used during the excavation of trenches. (5)

2.4 What is a *rodding eye*? (2)

2.5 What are the most effective gradients for drainpipes with the following diameters?

2.5.1 100ø (1)

2.5.2 150ø (1)

2.5.3 225ø (1)

2.6 Why is access to underground drains important? (2)

2.7 Anchor blocks are essential features in plumbing activities. Give TWO purposes of such blocks. (2)
[20]

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QUESTION 3: PLUMBING MAINTENANCE AND REPAIRS

3.1 List TWO options to repair big leaks on galvanised pipes. (2)

3.2 There are a number of steps that must be followed during the replacement of a thermostat. Name the first SEVEN (in sequential order). (7)

3.3 What determines the pressure of water in a house? (2)

3.4 Choose a colour code from COLUMN B that matches a pressure-relief valve in COLUMN A. Write only the letter (A–E) next to the question number (3.4.1–3.4.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.4.1	50 kPa	A	blue
3.4.2	100 kPa	B	green
3.4.3	200 kPa	C	yellow
3.4.4	400 kPa	D	red
3.4.5	600 kPa	E	black

(5 × 1) (5)

3.5 Name FOUR instruments that can be used for the purpose of removing obstructions or blockages from drainage systems. (4)

[20]

QUESTION 4: BOX GUTTERS AND CHIMNEY FLASHINGS

4.1 How would you best describe a *box gutter*? (2)

4.2 What is the purpose of flashing? (2)

4.3 Draw a cross-section sketch of abutment flashing. (4)

4.4 Which TWO tools are best suited to cut out GMS sheeting? (2)

4.5 Name THREE alternate materials that the NHBRC lists which can be used for box gutters. (3)

4.6 What is meant by the term *soakers*? (2)

[15]

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QUESTION 5: INSTALL A SOLAR WATER HEATER

5.1 What do we call the energy which is sent out by the sun? (1)

5.2 Give a brief description of the term "*solar energy*". (2)

5.3 How does the process of heating take place in a water solar system? List SIX facts. (6)

5.4 Name THREE types of active solar water-heating systems. (3)

5.5 Why are back-up systems important with solar water-heating? (2)

5.6 Differentiate between:

- *glazed flat-plate collectors* (2)
- *unglazed flat-plate collectors*. (2)

5.7 How would you describe the term "*demand water heater*"? (2)

5.8 Which part of an evacuated-tube collector absorbs solar energy? (1)

5.9 Why do integral collector storage systems (ICS) not work well in households especially in the mornings? (2)

5.10 List an important factor that must be kept in mind the during the installation of a solar heating system. (2)

[25]

QUESTION 6: BASIC BUSINESS CONCEPTS

6.1 Jimmy runs a business. He has R100 000 in the bank. He quotes for a big job that will last for FOUR months and says he will charge R200 000 for the job. The client accepts the quote. Jimmy needs to buy R30 000 worth of material.

Expenses:

Wages:	R10 000
Transport (Van):	R6 000
Rent:	R4 000

Indicate with a bar graph chart Jimmy's cash-flow projection. Clearly specify what each bar represents. (6)

6.2 Name TWO aspects of professional communication. (2)

6.3 Draw up TWO questions that an entrepreneur should ask when doing market research about potential customers. (2)

[10]

TOTAL: 100

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